

Security & Privacy: Federated Summary algorithm

Authors: F.C. Martin

Introduction

This document is intended to assess the risk of using the Federated Summary algorithm. The document is modelled after the guidelines for describing risks for a federated learning algorithm as described in the vantage6 Security & Privacy document [1].

The first section explains how the algorithm works and which data is shared between the parties. Note that we only discuss data that originates from the privacy sensitive database and not the data that is used by the vantage6 infrastructure. In the second part, we discuss which known federated learning vulnerabilities apply to this algorithm. Finally, we discuss how these vulnerabilities may be mitigated.

The algorithm

The summary algorithm is just that, it returns a summary of some dataset(s) much like what is present in R. Let D represent the culmination of dataset(s) where d_i is a column of D . Then the summary algorithm will return the following:

- Mean of d_i
- Standard deviation of d_i
- Number of missing rows of d_i
- Number of useable rows in d_i
- Range of d_i
 - o If d_i is categorical or factor data this will return a table of counts per unique category
 - o If d_i is numeric it returns the minima and maxima of the data

There are four types of parties involved in the algorithm; (1) The aggregator, (2) the data stations, (3) the client and (4) the vantage6 server [1]. We also present a flow diagram which explains the different steps of the algorithm which are then explained in the remainder of this section. Note that the server is not displayed as it merely acts as a communication hub between data station, aggregator and researcher.

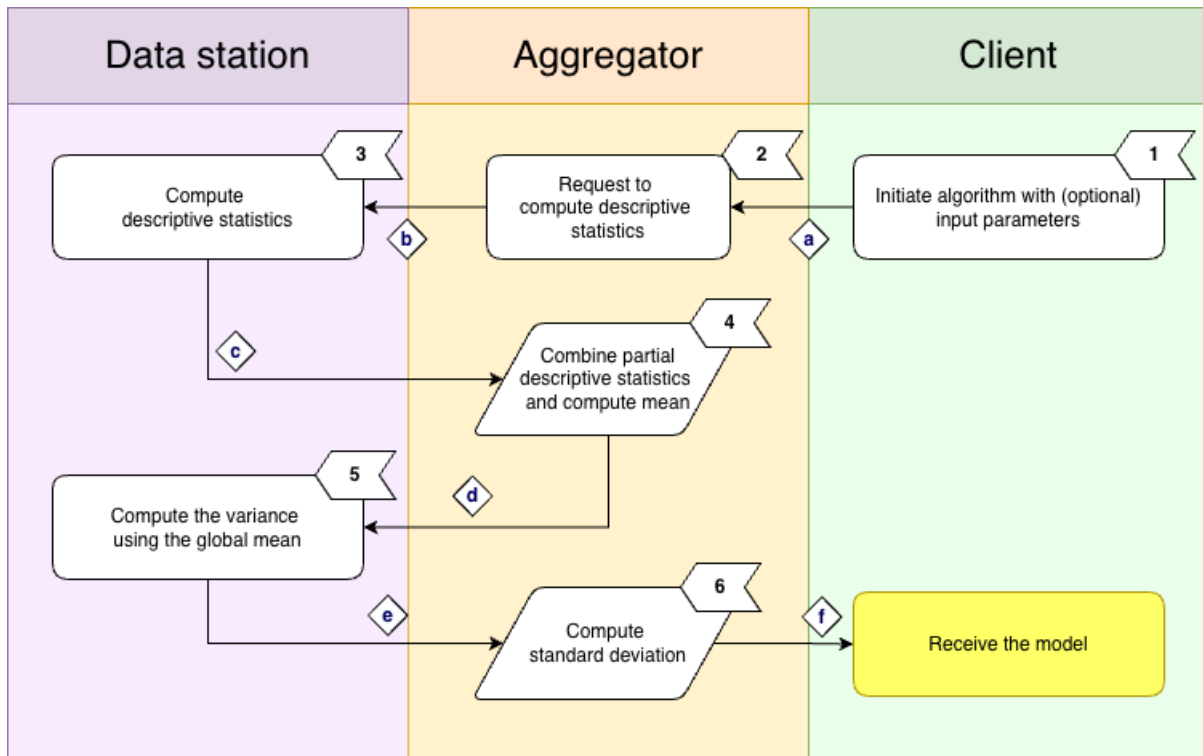


Figure 1 Schematic of the CoxPH algorithm. For a full description of the steps, see Section Algorithm Description. Note that all communication between these parties goes via Central Server.

Algorithm Description

The different steps of the algorithm are shown in Figure 1.

1. The researcher request to summarise a subset or all variables available in the dataset.
2. The aggregator request to compute the descriptive statistics for each local dataset.
3. Each data station computes the descriptive statistics. For numeric columns the min, max, sum and median is reported. For categorical columns all levels and their counts are reported. Finally, the overall count of records and number of records that do not have missing values is reported.
4. For the numeric columns we compute the global mean from the results reported in step (3), this mean is required in step (5) to compute the variance.
5. The variance is computed using the global mean.
6. Compute the global standard deviation from the reported variances.

Data in transit

Table 1 – A description of data transfers between vantage6 components. Note that all data transfers are mediated by the vantage6 server. The risk level comes from the original paper on Security and Privacy [1].

Description	Labels	Source	Destination	Risk
Column names	a, b	Client	Aggregator	Low
Descriptive statistics	c	Data station(s)	Aggregator	Low - High
Global mean	d	Aggregator	Data station(s)	Low
Variance	e	Data station(s)	Aggregator	Low – High
Descriptive statistics	f	Aggregator	Client	Low

As is indicated in the table above, the transferred data types that are potentially most sensitive are the *descriptive statistics* per node. However, whether these are sensitive in practice depends on the variables available and on the privacy settings.

Risks

In this section we will look at the types of attack and other kind of risks that the algorithm will be vulnerable to. Not all types of attack are relevant to this algorithm. Please refer to the Security and Privacy document [1] for the various types of attack definitions. It is important to note that the risk analysis is not exhaustive, and malicious parties will know more creative techniques.

Attack name	Risk analysis
Reconstruction	It is possible to reconstruct numerical variables in case the attacker has access to N-1 records, where N is the total number of records.
Differencing	This potentially is possible through the preprocessing steps but not from this algorithm itself.
Deep Leakage from Gradients (DLG)	Not applicable.
Generative Adversarial Networks (GAN)	Not applicable.
Model Inversion	Not applicable.
Watermark attacks	Not applicable.

Mitigation

Considerations to reduce the impact in the event of a successful attack.

- Differential privacy: There are methods for defending against this such as adding Gaussian noise [2]. A suitable way to protect against this without the use of Gaussian noise would be to simply require at least three data stations for an analysis. By adding Gaussian noise to the data, it effectively adds a veil over the data ensuring that even if the attacker manages to reconstruct/differentiate, what they receive back is some perturbed data and not the original.
- Parties > 2: Doing this ensures that even if there is a malicious party, data differencing and reconstruction becomes extremely difficult as it is tricky to discern from the aggregated statistics which data belongs to each member of the collaboration.
- Limit the available variables can reduce the risk when dealing with rare and sensitive data

References

- [1] Martin, F., van Gestel, A., van Swieten, M., Knoors D., van Beusekom, B., Geleijnse, G., 2023. 'Security and Privacy using vantage6 for Privacy Enhancing Analysis'.
- [2] Kairouz, P., Liu, Z. and Steinke, T., 2021, July. *The distributed discrete gaussian mechanism for federated learning with secure aggregation*. In *International Conference on Machine Learning* (pp. 5201-5212). PMLR.